

SilentMethyl: Multimodal Prediction of CpG Methylation and Single-Nucleotide Variant–Associated Methylation Changes, with Calibrated Uncertainty and Independent Cross-Tissue Variant Validation

Samuel Zhang^{1,2}, Shaojun Pei^{1,2,*}, and Gil Alterovitz^{1,2,*}

¹Division of General Internal Medicine and Primary Care, Brigham and Women’s Hospital, Boston, Massachusetts, United States

²Harvard Medical School, Harvard University, Boston, Massachusetts, United States

*Authors for correspondence: spei1@bwh.harvard.edu (Shaojun Pei); gil_alterovitz@hms.harvard.edu (Gil Alterovitz)

Abstract

DNA methylation is a central component of epigenetic regulation, yet directly measuring how individual nucleotide variants alter local methylation remains difficult because matched genetic and epigenetic data are scarce. We present SilentMethyl, a multimodal gated-fusion framework that combines a DNABERT-2 encoder over 1,000-bp CpG-centered sequence with an MLP encoding breast epigenomic context—chromatin accessibility, six histone marks, and evolutionary conservation. Learned gates weight the two representations per locus to predict CpG methylation, and paired wild-type/mutant scoring estimates variant-associated methylation change.

Under chromosome-blocked splitting, the fusion model reached a β -value MAE of 0.0993 and ROC-AUC of 0.9680 on 26,570 held-out CpGs, exceeding both single-modality baselines, with the largest gains at CpG shores and at loci with high accessibility or H3K27ac signal.

We evaluate variant effects on an independent cohort three orders of magnitude larger than prior work: 66,495 held-out variant–CpG pairs (39,657 unique variants) from GENOA, an African American whole-blood meQTL study, harmonised to hg38 and covering 19,081 of the 26,570 held-out probes. Among genome-wide significant associations the model recovered effect direction above chance (55.3%, 95% CI 53.6–57.0) and correlated with reported effects in signed rank ($\rho = 0.152$, 0.117–0.188). Agreement declined monotonically with association strength and reached chance in the null stratum, providing an internal negative control. Discrimination of true meQTLs from tested-but-null pairs survived matching on variant–CpG distance (AUROC 0.570, 0.554–0.585), though the unmatched estimate did not exceed a distance-only baseline—a comparison we report explicitly.

Scanning 831 JASPAR motifs across the held-out pairs recovers a specific regulatory relationship the model was never shown. Disruption of ETS-family motifs couples to predicted hypermethylation (ρ up to -0.28) against a null distribution of per-factor couplings centred at -0.004 , and survives control for local GC content, variant–CpG distance, and substitution class. Disruption magnitude alone predicts nothing, so what the model learned is signed and family-specific rather than a general sensitivity to motif damage.

Two further findings generalise beyond this model. First, uncertainty evaluation on bounded bimodal targets is confounded by range compression: a zero-parameter heuristic outperforms genuine uncertainty estimators on the β scale and collapses on the unbounded logit

scale, so calibration should be assessed in logit space. Second, gated multimodal fusion improves absolute methylation prediction but contributes nothing to variant-effect prediction (paired equivalence intervals bounded within ± 0.007), because context features are allele-invariant and therefore cannot create a variant effect. Together these results establish SilentMethyl as a calibrated framework linking sequence variation to local epigenetic regulation, and delimit what multimodal fusion can and cannot contribute to variant-effect prediction.

Draft status ^{NEW}

This section is draft scaffolding. Delete it, and set `\draftmodefalse`, before submission.

New since the original draft (main.tex). Section 2.4 and Section 3.4 (independent variant evaluation on GENOA); Section 2.5 (block bootstrap and distance-matched negatives); Section 2.6 and Section 3.6 (uncertainty quantification); Section 3.5 (fusion/sequence equivalence); Section 2.7 and Section 3.7 (TCGA-BRCA tumours as a shifted target domain); Section 2.8 and Section 3.8 (ETS motif disruption). Abstract, title, introduction, discussion and limitations revised throughout.

Open question inside a written section. The context-only model shows no degradation at all against the tumour target and its M MAE improves (Section 3.7). Genuine robustness of coarse chromatin signal and bounded-range compression from a less bimodal tumour β distribution both predict this. Resolve before the text asserts either; if it is compression, that is a third appearance of our own Section 3.6 finding in our own results and belongs in the discussion.

Not yet written, in priority order. (i) head-to-head against CpGenie and DeepCpG (repositories cloned, environments not yet stood up); (ii) repeated chromosome-blocked splits, four additional folds at seed 42—the only remaining model training; (iii) conformal prediction intervals; (iv) a ClinVar figure for clinical relevance; (v) HOCOMOCO as a second motif library, to confirm the ETS result is not specific to JASPAR’s matrices.

Bibliography. New citation keys required in `references.bib`: shang2023genoa, avsec2025alphagenome, ying2024methylogpt, gemma2012, castrocolin2013jaspar. If `references.bib` is absent at build time, this document falls back to

a key-only reference list so that it still compiles.

1 Introduction

DNA methylation is a widely studied epigenetic modification involved in gene regulation, associated with genomic context, chromatin organization, and transcription [1, 2], and frequently dysregulated in disease. Single-nucleotide variants (SNVs) can influence local epigenetic states, but validating their direct regulatory effects is difficult because matched genetic and epigenetic data are limited. This motivates modeling DNA methylation to identify potentially functional variants.

Existing deep learning methods predict DNA methylation and related regulatory measures from local DNA sequence, neighboring methylation, and long-range sequence context [3–5]. While DNA sequence contains motifs and local regulatory patterns, sequence alone misses the epigenetic context in which they operate. Reference epigenomic data, including chromatin accessibility, histone marks, and evolutionary conservation, can supply part of this missing information. Predicting methylation at a locus and estimating the effect of a base change on methylation are also distinct tasks: a model that predicts baseline methylation accurately may still estimate variant-induced changes poorly.

Several frameworks for evaluating paired sequence changes already exist. CpGenie predicts CpG methylation and validates REF–ALT sequence changes against mQTLs [6]. INTERACT focuses on brain methylation, DeepMethylation/DDM combines sequence with reference tissue data, and Methven combines sequence with ATAC-seq accessibility [7–9].

^{REV}A specific gap motivates this work. Recent long-context variant-effect models—Enformer, Borzoi and AlphaGenome—predict many regulatory modalities at megabase scale but do not predict CpG methylation [5, 10]. Conversely, methylation foundation models such as Methyl-GPT operate on measured methylation profiles rather than sequence, and so cannot score an unobserved variant at all [11]. Sequence-based prediction of *variant effects on methylation* therefore sits between two active literatures without being served by either.

Existing approaches also share two methodological limitations. Most quantify the contribution of sequence and epigenomic inputs only through aggregate model-performance differences, and most do not account for reverse-complement equivariance [12]. SilentMethyl addresses both: gated fusion exposes the relative weighting of sequence and context at each individual probe, and reverse-complement stability is enforced by random orientation during training and by averaging forward and reverse-complement predictions at inference, which makes the reported output exactly RC-invariant by construction.

SilentMethyl uses synonymous mutations as a focused baseline for variant prioritization, since they change a base without changing the encoded amino acid while still potentially affecting splice recognition, RNA structure, transcript

stability, translation, and exonic regulatory sequence [13, 14]. Several studies already link synonymous variants to DNA regulatory features: a synonymous *MLH1* c.27G>A allele was found in individuals with constitutional *MLH1* methylation [15]; synonymous *TP53* variants altering CpG content are linked to intragenic demethylation and alternative promoter usage [16]; synonymous *ADAMTS13* variants alter molecular properties by creating and destroying CpG sites [17].

^{REV}This work makes four contributions. (1) A gated multimodal architecture that improves CpG methylation prediction over either modality alone and exposes per-locus modality weighting. (2) Independent variant evaluation at a scale that supports stratified inference: 66,495 held-out variant–CpG pairs from an African American whole-blood cohort, evaluated against a distance-matched negative control with intervals that respect linkage disequilibrium. (3) A demonstration that uncertainty evaluation on bounded bimodal targets is confounded by range compression, with a scale-stability diagnostic that distinguishes genuine uncertainty signal from metric artifact. (4) A negative result with a mechanistic explanation: gated fusion cannot contribute to variant-effect prediction while its context features are allele-invariant. The last two apply to any sequence-plus-context model performing variant scoring.

2 Methods

2.1 Data construction

Methylation beta labels were constructed from the TCGA-BRCA HM450 matrix [18, 19]. Using a sample-type-11 filter, 97 solid-tissue-normal columns were selected from 893 total, verified to represent 97 unique participants without duplicate samples.

Single-probe methylation was represented as the median of patient beta values. For CpG i , if O_i is the set of patients with a recorded measurement at CpG i , then

$$\beta_i = \text{median}_{s \in O_i} \beta_{i,s}, \quad y_i = \mathbf{1}[\beta_i > 0.5],$$

$$\tilde{\beta}_i = \max\left(10^{-4}, \min\left(1 - 10^{-4}, \beta_i\right)\right),$$

$$M_i = \log_2\left(\frac{\tilde{\beta}_i}{1 - \tilde{\beta}_i}\right).$$

The beta-to- M transformation follows [20].

All probes required an exact CpG-centered DNA sequence and a valid reference context window. None of the 418,486 probes in the combined dataset were flagged by the HM450 MASK_general filter. Median probe coverage was 97 of 97, and 95.1% of loci had at least 78 valid patients. Probe coordinates and sequence were taken from the hg38 HM450 annotation resource [21, 22].

Data were split by chromosome before training: chromosomes 10–11 for validation, chromosomes 8–9 for test, remaining chromosomes for training (Table 1). Splitting by chromosome ensures no sequence windows overlap between

partitions.

Each DNA input was a 1,000-bp CpG-centered sequence with the target CpG nucleotides at zero-indexed positions 499 and 500. The epigenetic input comprised seven MCF-10A attributes—ATAC-seq accessibility and six histone marks (H3K4me3, H3K27ac, H3K27me3, H3K9me3, H3K36me3, H3K4me1)—averaged across a CpG-centered 100-bp window, plus two PhyloP scores for the two nucleotides of the central CpG. A missingness mask accompanied each feature; missing values were imputed with the feature’s global median. Reference data were derived from shared resources and were not patient-specific.

Table 1: Chromosome split dataset. Methylated means median $\beta > 0.5$.

Split	Loci	Mean β	Methylated
Train	345,359	0.4948	52.25%
Validation	46,557	0.5035	53.19%
Test	26,570	0.5029	53.20%

2.2 Models, training, and evaluation

The sequence module comprised a pretrained DNABERT-2 (117M) transformer [23] followed by a convolutional layer and masked attention pooling:

$$\alpha_t = \frac{\exp(s_t)}{\sum_{u \in V} \exp(s_u)}, \quad \mathbf{h}_{\text{seq}} = \sum_{t \in V} \alpha_t \mathbf{e}_t,$$

where V contains non-padding positions and $\mathbf{e}_t$ is the embedding at position t , producing a 768-dimensional $\mathbf{h}_{\text{seq}}$.

The context module was an MLP mapping the 18-dimensional epigenetic input (9 features + 9 missingness masks) to a 768-dimensional $\mathbf{h}_{\text{ctx}}$.

After LayerNorm produced $\tilde{\mathbf{h}}_{\text{seq}}$ and $\tilde{\mathbf{h}}_{\text{ctx}}$, the two were concatenated and passed through a gating MLP yielding scalars g_{seq} and g_{ctx} . The fused embedding was

$$\mathbf{z} = g_{\text{seq}} \tilde{\mathbf{h}}_{\text{seq}} + g_{\text{ctx}} \tilde{\mathbf{h}}_{\text{ctx}},$$

fed to two prediction heads. The regression head predicted M , converted by $\hat{\beta} = 2^{\hat{M}} / (1 + 2^{\hat{M}})$; the classification head predicted $\hat{y} = \mathbf{1}[\sigma(\text{logit}) > 0.5]$. The objective was Huber loss ($\delta = 1.345$) on regression plus binary cross-entropy on classification.

Sequence and context models were trained independently with AdamW, with DNABERT-2 fully fine-tuned. Sequence, context, and fusion training used up to 10, 15, and 10 epochs at learning rates 2×10^{-5} , 10^{-3} , and 5×10^{-4} . During fusion training the component encoders were frozen and only normalization, gating, and prediction modules were updated.

During training, forward or reverse-complement orientation was chosen with equal probability per probe. For seeds 42, 43, and 44 the checkpoint was selected by lowest validation beta MAE. Primary metrics were RMSE and MAE for β and M , and ROC-AUC for classification.

2.3 Paired sequence scoring and sSNV construction

For each SNV the predicted methylation difference was averaged across strands:

$$\Delta\hat{\beta} = \frac{(\hat{\beta}_{\text{MUT,FWD}} - \hat{\beta}_{\text{WT,FWD}}) + (\hat{\beta}_{\text{MUT,RC}} - \hat{\beta}_{\text{WT,RC}})}{2},$$

with $\Delta\hat{M}$ defined analogously. Because the reported prediction is $\frac{1}{2}[f(x) + f(\text{RC}(x))]$, it is exactly invariant under reverse complementation.

Tagged somatic sSNVs were obtained through the GDC with amino-acid preservation verified against hg38 and GENCODE v44 [19, 24]. Variants were matched to the nearest HM450 probe; 440 unique variants remained after removing MASK_general1-flagged probes and variants outside the 1,000-bp window. Variants were prioritized by $|\Delta\hat{\beta}|$ from the three-seed fusion ensemble.

2.4 Independent variant evaluation: GENOA ^{NEW}

External validation used cis-meQTL summary statistics from GENOA, an African American whole-blood cohort [25]. Three properties make it suitable: it is a different tissue, a different ancestry, and a different array generation from the training data, and it is openly released.

Harmonisation. Summary statistics are reported in hg19. The CpG was never lifted over: array probe identifiers are platform-stable, so the reported probe joins directly to the hg38 manifest and inherits hg38 coordinates. Only SNP positions required hg19→hg38 conversion. Of 942,186 candidate pairs, 731 (0.1%) had neither reported allele matching the hg38 reference base and were discarded; this rate is itself evidence that the liftover and strand convention are correct.

Effect-allele alignment. GENOA effect sizes are GEMMA output, in which β is the effect per copy of allele1, the minor allele [26]. The model’s prediction is the effect of substituting ALT for the hg38 reference. These directions coincide only when ALT is the minor allele. In 106,628 pairs (11.3%) the hg38 reference base is the minor allele, so ALT is the major allele and the two conventions are opposed. All comparisons therefore use the published effect re-signed to the REF→ALT direction. This correction is silent if omitted—it does not produce an error—and it attenuates both signed correlation and direction agreement.

Strata. Pairs were labelled by the split membership of their target probe, read from the actual training partitions rather than inferred from chromosome number. Only the held-out stratum supports an independent-validation claim: 66,495 pairs, 39,657 unique variants, 19,081 of the 26,570 held-out probes. Pairs whose variant falls at the target CpG itself were excluded (the architecture is defined around a CpG at window

Figure workflow.pdf not present in this build.

Figure 1: SilentMethyl data construction, model architecture, and evaluation workflow. Panel A shows methylation target construction, chromosome partitioning, and model inputs. Panel B shows FWD-RC orientation and staged training. Panel C shows the gated multimodal architecture. Panel D shows paired WT-MUT scoring, held-out probe testing, external mQTL validation, and synonymous-variant ranking.

indices 499:501, and such array measurements are a known SNP-under-probe artifact), as were pairs outside the 1,000-bp window, for which the model returns a delta of exactly zero. Variants that create or destroy a CpG dinucleotide have a near-deterministic direction requiring no model and are reported as a separate stratum throughout; unless stated otherwise, results below refer to the 42,866 non-CpG-altering held-out pairs.

2.5 Statistical framework ^{NEW}

Dependence. The held-out pairs comprise 39,657 variants in linkage disequilibrium across 19,081 probes and are far from independent. All intervals are percentile bootstrap intervals resampling whole 1-Mb genomic blocks rather than individual pairs; row-level resampling would treat correlated observations as independent and yield intervals several times too narrow.

Distance matching. Significant meQTLs lie closer to their target CpG than tested-but-null pairs (median 191 versus 258 bp), and predicted effect magnitude increases with proximity ($\rho = -0.29$ between $|\Delta M|$ and distance). Variant-CpG distance alone therefore separates the two classes, and an unmatched discrimination estimate partly measures distance rather than regulatory effect. Each significant pair was matched to a null pair at the same base-pair distance, sampled without replacement so that no null could be reused. Within the matched cohort, distance alone achieves AUROC 0.500 by construction, so residual discrimination is attributable to the model. The distance-only baseline is reported alongside every discrimination estimate.

2.6 Uncertainty quantification ^{NEW}

Averaging forward and reverse-complement predictions makes the output invariant but discards the disagreement between them. That disagreement is available at every locus at

no additional cost and was evaluated as a single-model uncertainty estimate against a three-seed ensemble standard deviation, a zero-parameter boundary heuristic $-\lvert\hat{\beta} - 0.5\rvert$, and a random control, using risk-coverage curves and area under the risk-coverage curve.

Because β is bounded on $[0, 1]$ and bimodal, absolute β error is mechanically compressed near the boundaries. Every comparison was therefore repeated on the unbounded M scale, and additionally within strata of predicted β at 10, 20, and 50 levels, so that an estimator’s apparent skill could be separated from headroom in the target.

2.7 Tumour cohort as a shifted target domain ^{NEW}

The TCGA-BRCA matrix already used for training contains 791 primary-tumour columns that were never used. Of these, 699 come from participants absent from the 97 solid-tissue-normal samples used to build the training targets; those 699 form the primary cohort, and the full 791 are reported as a supplement because the remaining 91 participants make that comparison paired rather than independent.

One property of this design must be stated plainly. Model inputs are the reference sequence and MCF-10A context features, neither of which depends on disease state, so *predictions are identical for normal and tumour*. This evaluation holds predictions fixed and swaps the target. It therefore measures the component of the tumour methylome determined by sequence and reference chromatin context, and where the model degrades is where methylation has been reprogrammed by something the model cannot observe. It is not, and is not presented as, generalisation to a new input domain.

2.8 Motif disruption analysis ^{NEW}

JASPAR CORE vertebrates [27] (877 matrices, of which 831 met a 200-variant coverage threshold) was scanned across the 42,866 non-CpG-altering held-out GENOA pairs. Counts

were converted to log-odds matrices against a uniform background with a Laplace pseudocount, and scores were expressed relative to each matrix’s own range so that a threshold means the same thing across motifs of different length and information content; 0.80 is the conventional cutoff.

Both strands were scanned, since motifs are strand-specific. For each pair, the best-scoring wild-type occurrence covering the variant was located, and the mutant was scored *at that same offset and strand*: choosing the optimum separately per allele would allow apparent disruption to be the motif relocating to a different site. Per-factor coupling is the Spearman correlation between the change in relative motif score and the predicted change in M , across pairs covered by that motif, with Benjamini–Hochberg correction across factors.

Three controls were applied, each targeting a distinct way the result could be an artifact. A *null distribution* across all tested motifs distinguishes a factor-specific effect from a global offset in which disrupting any motif shifts predictions one way. *Partial correlations* residualise local GC content and variant–CpG distance, since motifs concentrate in GC-rich sequence and GC-rich sequence sits nearer CpG islands. A further partial adds *canonical substitution class* (six categories, pyrimidine reference), because compositionally biased motifs imply biased base changes: weakening a purine-rich motif usually means a G/A→C/T substitution, and pyrimidine-rich sequence is more methylated on average.

Because motif families share a core, top-ranked factors are not independent observations. Pairwise Jaccard overlap of the covered-variant sets is reported so that redundancy is measured rather than assumed.

A binary inside-versus-outside-a-motif contrast is not usable here: with the full library scanned, every variant falls inside some occurrence at the 0.80 cutoff, leaving no background. Contrasts therefore use top-versus-bottom quartiles of disruption magnitude, matched on exact variant–CpG distance and GC quintile.

2.9 External eGTEx mQTL analysis

Significant eGTEx Breast Mammary Tissue lead mQTLs ($q < 0.05$) were overlapped with test-set CpGs [28]. After excluding variants outside the sequence window or altering their target CpG, 81 associations with 70 unique variants remained. Predicted ALT–REF changes were compared with reported allele slopes using signed and absolute Spearman correlation, sign agreement, and AUROC. A sensitivity analysis dropped mutations within or immediately adjacent to a probe. A matched analysis tested whether predicted magnitude separated significant leads from high- q associations ($q \geq 0.5$) matched on chromosome and minor allele frequency (35 pairs from 53 significant and 51 high- q variants).

Figure model_incremental_performance.png
not present in this build.

Figure 2: Held-out chromosome performance. Gray lines connect the same training seed. (A) M-value MAE. (B) beta MAE. (C) ROC-AUC.

Table 2: Held-out chr8–9 performance. Mean $\pm$ SD across three seeds on 26,570 CpGs.

Metric	Context	Sequence	Fusion
M-value MAE	1.4574 $\pm$ 0.0023	1.1941 $\pm$ 0.0070	1.0971 $\pm$ 0.0192
Beta MAE	0.1395 $\pm$ 0.0001	0.1099 $\pm$ 0.0008	0.0993 $\pm$ 0.0025
ROC-AUC	0.9187 $\pm$ 0.0001	0.9569 $\pm$ 0.0007	0.9680 $\pm$ 0.0021

3 Results

3.1 Multimodal fusion outperforms single-modality models

On the 26,570-probe test set the fusion model was strongest across all metrics and seeds. The context MLP established a baseline beta MAE of 0.1395 ± 0.0001 and ROC-AUC 0.9187 ± 0.0001 . The sequence model reduced beta MAE by 21.2% (0.1099 ± 0.0008) and raised ROC-AUC to 0.9569 ± 0.0007 . Fusion improved beta MAE by a further 9.6% (0.0993 ± 0.0025) and raised ROC-AUC to 0.9680 ± 0.0021 (Fig. 2). Predictions were stable at lower coverage: on probes measured in at least 78 of 97 patients the seed-42 fusion model reached beta MAE 0.09669 and ROC-AUC 0.97066.

Improvement was consistent across seeds and across genomic regions. Fusion improved sequence-only MAE by 0.01043 (95% CI 0.00951–0.01132) and ROC-AUC by 0.01053 (0.00909–0.01207) under a 257-block resampling design.

3.2 Reference context contributes across regulatory conditions

Fusion reduced beta MAE in every stratum of GENCODE region, CpG island status, ATAC-seq quartile, and H3K27ac quartile. By GENCODE annotation the reduction ranged from 0.0088 at intergenic probes to 0.0130 at UTRs. By island status the largest gain was at shores (0.0162). Fusion low-

Figure fusion_gain_by_epigenomic_context.png
not present in this build.

Figure 3: Fusion gain across epigenomic contexts. Intervals re-sample 1-Mb regions within each stratum. (A) CpG island status. (B) ATAC-seq quartile. (C) H3K27ac quartile.

ered beta MAE by 0.0139 in the highest ATAC-seq quartile and 0.0147 in the highest H3K27ac quartile (Fig. 3). Context therefore contributes most where regulatory state is expected to carry additional information.

3.3 Breast mQTLs support predicted effect direction

On 81 independent breast mQTL associations the three-seed fusion ensemble recovered 64 of 81 effect directions (79.0%), achieved signed Spearman $\rho = 0.610$, and separated positive from negative slopes with AUROC 0.872. The pattern held after removing variants on or adjacent to a probe (53 associations; $\rho = 0.610$, 75.5% direction agreement, AUROC 0.872).

Absolute magnitudes agreed far less well ($\rho = 0.217$ overall, 0.282 after filtering), and the model did not separate matched significant leads from high- q leads on $|\Delta\hat{M}|$ (AUROC 0.512, 95% CI 0.380–0.646). At $n = 81$ these estimates are imprecise, which motivated the larger independent evaluation below.

3.4 Independent variant evaluation at scale ^{NEW}

Scoring 66,495 held-out GENOA pairs increased the independent variant evaluation from 81 associations to 39,657 unique variants, a scale that supports stratified inference rather than a single pooled estimate (Table 3).

Effect direction was recovered above chance and signed rank correlated with reported effects, with both intervals excluding the null. Effect sizes are modest: $\rho \approx 0.15$ and 55% direction agreement describe a weak predictor, and we report it as such. What the analysis establishes is that the effect is real and measurable on an independent cohort differing in tissue, ancestry, and platform generation, on probes the model never saw.

Table 3: Independent variant evaluation on GENOA. Held-out stratum, non-CpG-altering pairs, three-seed ensemble. Brackets are 95% intervals from a block bootstrap over 1-Mb blocks. Signed ρ and direction agreement are computed on genome-wide significant pairs ($p < 5 \times 10^{-8}$, $n = 4,037$); AUROC uses all 42,866 pairs.

Metric	Fusion	Sequence
Signed ρ	+0.152 [+0.117, +0.188]	+0.150 [+0.109, +0.184]
Direction agreement	0.553 [0.536, 0.570]	0.554 [0.538, 0.573]
AUROC, distance-matched	0.570 [0.554, 0.585]	0.562 [0.546, 0.578]
AUROC, unmatched	0.600 [0.586, 0.614]	0.603 [0.589, 0.617]
AUROC, distance only	0.595 [0.583, 0.609]	

Table 4: Agreement by association strength. Fusion, three-seed ensemble, non-CpG-altering held-out pairs.

Stratum	n	Signed ρ	Direction
$p < 5 \times 10^{-8}$	4,037	+0.152 [+0.117, +0.184]	0.553
0.05–0.5	15,891	+0.023 [+0.006, +0.040]	0.508
$p > 0.5$	13,330	−0.004 [−0.020, +0.014]	0.497

Figure significance_gradient.png
not present in this build.

Figure 4: Agreement with GENOA rises with association strength and reaches chance where associations are null. Signed Spearman correlation (left) and direction agreement (right) across seven strata of reported p -value, with 95% block-bootstrap intervals.

An internal negative control. GENOA publishes every cis pair tested rather than only the meQTLs discovered, and 71% of held-out non-CpG-altering pairs have $p > 0.05$. Stratifying by association strength shows agreement rising with the strength of the underlying association and reaching chance where no association exists (Table 4, Fig. 4). The null stratum is statistically indistinguishable from zero on both metrics, providing a negative control internal to the same dataset. Intermediate strata are not strictly monotone and their intervals overlap; the interpretable contrast is significant versus null.

Discrimination is not explained by distance alone. Variant–CpG distance by itself separates true meQTLs from null pairs at AUROC 0.595 (0.583–0.609), statistically indistinguishable from the model’s unmatched 0.600 (0.586–0.614). We state this explicitly: *marginally, predicted effect magnitude does not outperform distance*. After matching each significant pair to a null pair at identical distance, however, discrimination remains at 0.570 (0.554–0.585), and exceeds

Figure discrimination_vs_distance.png
not present in this build.

Figure 5: Discrimination survives distance matching. AUROC for separating genome-wide significant meQTLs from tested-but-null pairs, unmatched and within distance strata, against the distance-only baseline.

chance in every distance bin examined. The model’s signal is therefore largely orthogonal to distance rather than a restatement of it (Fig. 5).

3.5 Gated fusion does not contribute to variant-effect prediction ^{NEW}

Fusion and sequence-only models were statistically equivalent on every variant-effect metric. Paired differences, bootstrapped over the same genomic blocks, were +0.0026 (−0.0020, +0.0070) for signed ρ , −0.0007 (−0.0064, +0.0051) for direction agreement, and −0.0021 (−0.0053, +0.0011) for within-distance-bin AUROC. The intervals are narrow enough to support equivalence rather than merely a failure to detect a difference: any difference is bounded within ± 0.007 .

This follows from the architecture. The context vector depends on the probe, not on the allele, and is therefore identical for wild-type and mutant input. Writing the fused embedding difference,

$$\Delta \mathbf{z} = [\tilde{\mathbf{h}}_{\text{seq}}^{\text{mut}} g_{\text{seq}}^{\text{mut}} - \tilde{\mathbf{h}}_{\text{seq}}^{\text{wt}} g_{\text{seq}}^{\text{wt}}] + \tilde{\mathbf{h}}_{\text{ctx}} [g_{\text{ctx}}^{\text{mut}} - g_{\text{ctx}}^{\text{wt}}],$$

the context term survives only through a gate shift that is itself driven by the sequence change. Context can modulate a variant effect; it cannot create one.

The gate is the only route by which allele-invariant context could still shape a prediction, and that route is also closed: the fusion-minus-sequence difference is flat across quartiles of gate DNA share, with all eight intervals covering zero and the largest point estimate 0.007. A faint sign pattern runs in the predicted direction across quartiles but is not statistically supported.

This is consistent with the earlier observation that fusion and sequence rankings of the 440 synonymous candidates were highly concordant (signed $\rho = 0.981$, 96.6% sign agreement). The present analysis explains why, and bounds the effect.

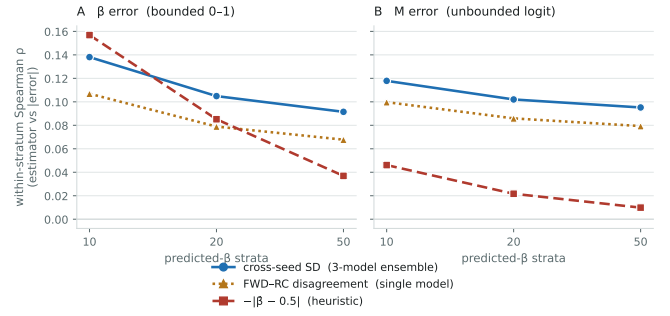


Figure 6: Scale stability separates uncertainty signal from metric artifact. Within-stratum rank correlation between each uncertainty estimator and absolute error, on the bounded β scale and the unbounded M scale. Genuine estimators are scale-stable; the boundary heuristic is not.

3.6 Uncertainty is quantifiable, but the standard metric is confounded ^{NEW}

Forward/reverse-complement disagreement tracked per-locus error at Spearman $\rho \approx 0.50$ for the sequence and fusion models (0.35–0.39 for context-only), consistently across seeds. On selective prediction, however, the zero-parameter heuristic $-|\hat{\beta} - 0.5|$ outperformed every genuine estimator—including the three-seed ensemble—in 8 of 9 model–seed combinations.

That result is an artifact of the target rather than a property of the estimators. Absolute β error is mechanically compressed near 0 and 1, so a model that is merely confident at extreme predictions appears well calibrated. Repeating the comparison on the unbounded M scale, and stratifying finely by predicted β , separates the two explanations: the heuristic’s within-stratum correlation collapses from +0.0369 to +0.0099 between the β and M scales, while cross-seed ensemble standard deviation (+0.0915 to +0.0952) and FWD–RC disagreement (+0.0678 to +0.0793) are unchanged (Fig. 6). At 50 strata both genuine estimators outperform the heuristic on both scales.

Scale-instability is the signature of a metric artifact. A genuine uncertainty signal ranks error about equally well on either scale; a compression artifact does not. We recommend that calibration for bounded bimodal targets be assessed in logit space, and note that β MAE is the field’s routine reporting choice.

FWD–RC disagreement retains 75–83% of the ensemble’s incremental information depending on the measure, from a single model at no additional inference cost.

3.7 Tumour methylation is predicted from sequence where it has not been reprogrammed ^{NEW}

Holding predictions fixed and scoring them against the 699-tumour target, fusion β MAE moved from 0.0975 to 0.1180 and ROC-AUC from 0.9696 to 0.9448; the sequence model behaved similarly (0.1079 to 0.1296; 0.9590 to 0.9308). Sixty-five per cent of held-out probes differ by less than 0.05 be-

Table 5: Strongest per-factor coupling between motif disruption and predicted methylation change. Spearman ρ between the change in relative motif score and $\Delta\hat{M}$, among held-out non-CpG-altering pairs covered by each motif. Successive columns add controls. All $q_{BH} < 10^{-12}$.

Factor	n	ρ	+GC/dist	+subst.	motif GC
ELF4	1,005	-0.266	-0.268	-0.281	0.55
FEV	2,068	-0.257	-0.259	-0.257	0.56
EHF	1,711	-0.251	-0.253	-0.252	0.51
ELF1	1,352	-0.242	-0.246	-0.233	0.54
ETV1	1,606	-0.237	-0.242	-0.240	0.49
GABPA	968	-0.233	-0.235	-0.230	0.50

tween the normal and tumour targets, and Spearman correlation between the two targets is 0.956.

Degradation is concentrated exactly where the methylome moved: fusion β MAE against the tumour target rises monotonically across strata of $|\beta_{\text{tumour}} - \beta_{\text{normal}}|$, from 0.0769 in the smallest stratum to 0.4082 in the largest. Most of the tumour methylome at these probes is therefore recoverable from sequence and reference chromatin context alone, and the model’s error is an estimate of the state-specific remainder.

The context-only model is a partial exception: its β MAE is unchanged (0.1395 versus 0.1396) and its M MAE improves. Whether this reflects genuine robustness of coarse chromatin signal or the bounded-range compression described in Section 3.6 reappearing because tumour β is less bimodal is not yet resolved (see draft status).

3.8 Disrupting ETS-family motifs predicts hypermethylation ^{NEW}

Across 831 tested motifs, the strongest couplings between motif score change and predicted methylation change form a coherent group of ETS-family factors, all with the same sign (Table 5). Negative coupling means motif weakening accompanied by predicted *hypermethylation*, the direction expected for factors whose binding protects CpGs from methylation.

The result survives three controls, each addressing a distinct alternative explanation.

It is not a global offset. Median coupling across all 831 motifs is -0.0037 (IQR -0.045 to $+0.038$), 51.4% of motifs are negative, and 185 are significantly negative against 155 significantly positive at $q < 0.05$. The null is centred and symmetric, placing the ETS group roughly five interquartile ranges outside it.

It is not GC or distance. Partial correlations are unchanged or slightly stronger. A library-wide gradient exists ($\rho = -0.20$ between motif GC and coupling), but the ETS motifs are mid-GC (0.44–0.65); were composition driving the result, GC-rich motifs at 0.85–0.90 would rank highest, and they do not.

It is not substitution bias. ETS cores are purine-rich, so weakening one typically implies a G/A→C/T change, and pyrimidine-rich sequence is more methylated on average. Residualising on canonical substitution class leaves every cou-

Figure motif_disruption.png
not present in this build.

Figure 7: Motif disruption and predicted variant effects. Left: per-factor coupling between change in relative motif score and $\Delta\hat{M}$, with 95% block-bootstrap intervals; asterisks mark factors with previously reported methylation-sensitive binding. Right: meQTL discrimination among strongly versus weakly disrupting variants, which does not differ.

pling essentially unchanged.

Two null results accompany the positive one and are reported with it. The *magnitude* of disruption predicts nothing (continuous $\rho = -0.011$, -0.023 to $+0.002$; strong versus weak median $|\Delta\hat{M}|$ 0.0177 versus 0.0184), and meQTL discrimination is not concentrated among strong disruptors (AUROC 0.592, 0.570–0.614, versus 0.596, 0.574–0.620). The model did not learn that disrupting motifs matters in general; it learned a signed, family-specific relationship.

Two constraints on interpretation. First, the top-ranked factors are not independent: mean pairwise Jaccard overlap of their covered-variant sets is 0.278 and the maximum is 0.792, so this is the ETS core motif represented by several matrices rather than several independent findings. Second, the 18 factors with previously reported methylation-sensitive binding have median rank 136 of 831 – enriched relative to a chance expectation of 416, but decisively outranked by ETS.

3.9 Variant prioritization and case studies

Applied as a prioritization engine to 440 synonymous variant–CpG pairs, the fusion model produced consistent rankings across seeds ($r = 0.681$ – 0.699). Median $|\Delta\hat{\beta}|$ was 0.00360 at promoter/TSS variants, 0.00261 at UTRs, and 0.00165 in gene bodies; variants within 50 bp of the target CpG had median 0.00512 against 0.00111 at 251–500 bp (Fig. 8). Neither genomic region nor distance was supplied as a feature.

The highest-ranked synonymous candidate, chr8:g.70148394G>A in *NCOA2*, is a GGC>GGT substitution 9 bp from probe cg20699548, which lies in an OpenSea context in the top quartile of both ATAC-seq and H3K27ac signal. The ensemble predicted a hypomethylating shift of $\Delta\beta = -0.1798 \pm 0.0198$, exceeding all 66 matched synonymous background variants (Fig. 9). *NCOA2* encodes a nuclear receptor coactivator with reported involvement in breast tumour growth [29]; no published evidence links this specific

Figure candidate_response_by_context.png
not present in this build.

Figure 8: Candidate response by biological context. Fusion difference magnitude grouped by (A) variant region and (B) distance to CpG.

variant to *NCOA2* activation, and the prediction constitutes a testable hypothesis rather than a finding.

Figure top_candidate_matched_background.png
not present in this build.

Figure 9: Top candidate versus matched background. The absolute response of the *NCOA2*-associated candidate compared with 66 matched observed synonymous variants.

The framework extends to any single-nucleotide change. Among studied nonsynonymous variants in breast-cancer genes, the two highest-ranked were both in *STK11* [30]: rs2145420809 ($\Delta\hat{\beta} = 0.0774$ at cg16601904, immediately adjacent) and rs148928808 ($\Delta\hat{\beta} = -0.0613$ at cg08681293, 204 bp downstream), the latter a known pathogenic truncating variant. Both target CpGs lie on training chromosomes, so this is an illustrative application rather than validation.

4 Discussion

SilentMethyl unifies two tasks in regulatory genomics: predicting baseline CpG methylation and estimating how that state responds to single-nucleotide variation. Integrating local sequence with breast epigenomic context consistently improved methylation prediction over either modality alone, with the largest gains at CpG shores and at loci with high accessibility or H3K27ac signal—regions where methylation is ex-

pected to reflect an interaction between sequence and chromatin state rather than sequence alone [1, 2, 31].

^{REV}The independent variant evaluation is the central addition of this revision, and it supports a specific and bounded claim. On 39,657 variants from a cohort differing in tissue, ancestry and platform generation, predicted effects recovered direction above chance and correlated with reported effects in signed rank, and discrimination of true meQTLs from tested-but-null pairs survived matching on variant–CpG distance. The effect sizes are modest and we do not present them otherwise. Their value lies in three properties the earlier 81-association analysis could not provide: the scale to stratify rather than pool, an internal negative control in the form of a null stratum that returns to chance, and intervals that respect linkage disequilibrium.

The distance analysis deserves emphasis because it is the most likely source of an inflated claim in this literature. Predicted effect magnitude is strongly related to variant–CpG proximity, true meQTLs are closer to their CpGs than null pairs, and distance alone therefore reaches AUROC 0.595—indistinguishable from the model’s unmatched estimate. Reporting the unmatched number in isolation would substantially overstate what the model contributes. We report it alongside the baseline and alongside the distance-matched estimate, which is the quantity that reflects the model’s own information.

^{REV}Two findings generalise past this model. The first is methodological: uncertainty evaluation on bounded bimodal targets is confounded by range compression, so that a zero-parameter heuristic can outperform genuine uncertainty estimators on the β scale while collapsing on the logit scale. Because β MAE is the standard reporting choice in methylation prediction, any model that is simply confident at extreme predictions will appear well calibrated under current practice. Scale stability provides a cheap diagnostic, and logit-space evaluation a straightforward remedy.

The second is architectural. Gated multimodal fusion improved absolute methylation prediction substantially but contributed nothing measurable to variant-effect prediction, with paired equivalence intervals bounded within ± 0.007 and no rescue through the gate. The explanation is structural rather than empirical: the context vector is identical for reference and alternate input, so it cannot create a variant effect. The implication for the field is concrete—adding richer or better-matched epigenomic context to a model of this form will not improve variant-effect prediction, whereas making context features themselves allele-dependent might. Notably, this also means that the mismatch between the breast context features used here and the blood tissue of the validation cohort is not what limits variant-effect performance.

^{REV}The motif analysis supplies the mechanistic layer the earlier draft lacked. A model given no motif information, trained only to predict methylation, recovered a signed and family-specific relationship: weakening an ETS-family binding site accompanies predicted hypermethylation. ETS and GABPA occupancy is an established hallmark of unmethylated CpG-island promoters, so the biology itself is not new.

What is new is the route — unsupervised recovery from sequence alone, detectable on held-out probes in a different tissue, ancestry and platform generation — and the quantitative ordering, in which previously characterised methylation-sensitive factors are enriched relative to chance yet decisively outranked. We state the distinction explicitly because the alternative reading, that this constitutes a new regulatory discovery, would not survive scrutiny and is not what the data support.

The accompanying null results constrain the claim usefully. Disruption magnitude does not predict effect size, and meQTL discrimination is no higher among strong disruptors. A model that had learned a general sensitivity to motif damage would show both. That neither appears, while a signed family-level coupling does, is more informative than the positive result alone, and is the reason we report the three together.

Limitations

Methylation targets derive from cohort medians of 97 TCGA solid-tissue-normal breast samples collected adjacent to tumours, which may retain field effects or cell-composition heterogeneity [18, 32]. Epigenomic context comes from MCF-10A reference tracks rather than matched samples, and HM450 covers only a subset of genomic CpGs.

The external validation cohort is whole blood while the model’s context features are breast; results on GENOA are therefore cross-tissue transfer rather than tissue-matched validation. GENOA effect sizes are reported on a normalized-phenotype scale, so rank and sign comparisons are meaningful while magnitude calibration against them is not available.

Motif occurrences are predicted from position weight matrices, not measured binding; without matched ChIP-seq in the relevant tissue these remain candidate sites. The top-ranked factors share a core motif and are not independent observations (mean pairwise Jaccard overlap 0.278 of their covered-variant sets), so the finding concerns one motif family rather than a set of factors. The tumour evaluation holds model inputs fixed and varies only the target, and measures the sequence-determined component of the tumour methylome rather than generalisation to a new input domain.

Ancestry-stratified error on the training cohort is not adequately powered: open genetic ancestry calls for the 97 training normals yield 84 European, 4 African, and 1 South Asian participant. Cross-ancestry evidence therefore rests on the contrast between the African American GENOA cohort and European-dominant reference data, which also differ in tissue and so do not constitute a clean ancestry contrast.

Evaluation currently uses a single chromosome-holdout design; repeated chromosome-blocked splits, head-to-head comparison against CpGenie and DeepCpG, regulatory-enrichment analysis, multi-cohort evaluation on unpaired TCGA-BRCA tumours, and conformal prediction intervals are in progress and are listed in the draft-status section.

5 Data and Code Availability

Code, run instructions, and Supplementary Data are available at <https://github.com/ilikecheezits/SilentMethyl>. Primary sources are the GDC, eGTE_x, GENOA, ENCODE, GENCODE, UCSC, and the HM450 annotation resource [19, 21, 24, 25, 28, 31, 33]. Harmonised GENOA variant–CpG pairs, per-seed variant scores, and the evaluation outputs underlying Tables 3 and 4 are included in the Supplementary Data.

6 Conclusion

SilentMethyl predicts CpG methylation and paired methylation differences caused by single-nucleotide variation in unseen chromosome regions. Fusion outperformed single-modality baselines for locus-level prediction. On an independent cohort of 39,657 variants differing in tissue, ancestry, and platform, predicted variant effects agreed with reported effects in direction and rank, with discrimination that survives matching on variant–CpG distance. Two accompanying findings delimit the approach: uncertainty on bounded bimodal targets must be assessed in logit space, and gated fusion cannot contribute to variant-effect prediction while its context features remain allele-invariant.

References

references.bib was not available when this draft was built, so citation keys are shown in place of formatted references. Building alongside the project bibliography restores the full list. Keys marked † are new in this revision and must be added.

References

- [1] jones2012functions
- [2] schubeler2015function
- [3] angermueller2017deepcpG
- [4] zhou2015predicting
- [5] avsec2021effective
- [6] zeng2017cpGgenie
- [7] zhou2022interact
- [8] li2026deepmethylation
- [9] liu2025methven
- [10] avsec2025alphagenome †
- [11] ying2024methy1gpt †
- [12] mallet2021rc
- [13] sauna2011understanding
- [14] stergachis2013exonic
- [15] alvarez2025mlh1
- [16] blackburn2016tp53
- [17] jankowska2022adamts13
- [18] tcga2012breast
- [19] grossman2016gdc
- [20] du2010betamvalue
- [21] zhou2017infinium
- [22] chen2013crossreactive

- [23] zhou2024dnabert2
- [24] frankish2021gencode
- [25] shang2023genoa †
- [26] gemma2012 †
- [27] castrocolin2013jaspar †
- [28] oliva2023mqt1
- [29] cai2019ncoa2
- [30] lipsa2019stk11
- [31] encode2012integrated
- [32] teschendorff2016field
- [33] kent2002ucsc